

Mario Avolio

AI & CLOUD ENGINEER · GENERATIVE & AGENTIC AI

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Summary

AI & Cloud Engineer focused on production-grade generative AI: LLM applications, agentic workflows, and RAG pipelines, served as containerized microservices on the cloud. Research background in computer vision and model optimization, with a Best Paper Award at ICIAP 2025. I build reliable, evaluable systems that move from prototype to production.

Experience

Machine Learning Reply

Milan, Italy

MACHINE LEARNING ENGINEER

Jan. 2026 - Present

- Build agentic workflows, RAG pipelines, and LLM-powered automation for enterprise use cases, from prototyping through deployment.
- Deploy and operate generative-AI services on Azure cloud infrastructure with containerized microservices and GitLab CI/CD pipelines.
- Apply evaluation and observability practices to keep AI systems reliable once exposed to real users and data.

Imaging and Vision Laboratory (IVL), University of Milano-Bicocca

Milan, Italy

RESEARCH FELLOW

Jan. 2025 - Dec. 2025

- Built a large-scale synthetic Side-Scan Sonar dataset of 20,000+ labeled images for underwater mine detection, beyond the scale of existing public datasets.
- Applied and optimized super-resolution models (SPAN, DRCT) to recover detection performance comparable to high-resolution data.
- Trained lightweight YOLOv11n/s detectors for robust detection in low-visibility conditions; work received a Best Paper Award at ICIAP 2025.

Skills

Generative AI & LLMs	LLM applications, RAG, multi-agent / agentic workflows, tool-calling, MCP, prompt & evaluation design, fine-tuning (LoRA/PEFT), vector databases
Frameworks & Tooling	LangChain, CrewAI, OpenAI Agents SDK, Microsoft Agent Framework, Hugging Face, Ollama, ChromaDB, Pydantic
Applied ML & Optimization	PyTorch, model optimization for cost-efficient inference (quantization, pruning, distillation), computer vision
Cloud & MLOps	Azure, Docker, FastAPI, CI/CD (GitLab, GitHub Actions), uv, pytest, model serving & observability, Linux
Programming	Python, SQL, Bash, Java, C++
Languages	Italian (native), English (fluent)

Projects

RAG · Multi-agent · Cloud

AI MICROSERVICES HUB

github.com/MarioAvolio/Portfolio

- A FastAPI gateway fronting independent generative-AI microservices: a profile RAG assistant (LangChain + ChromaDB), multi-agent research (OpenAI Agents SDK), and CrewAI-based competitive intelligence, with pluggable LLM providers (OpenAI, Gemini, Hugging Face, local Ollama). Gateway adds request-ID tracing, a bounded audit trail, and an optional API-key guard for production-style observability; each service containerized with its own tests and CI (Python, uv, Docker Compose).

Education

University of Milano Bicocca

Milan (MI), Italy

M.S. IN COMPUTER SCIENCE (GRADE: 110/110 CUM LAUDE)

Sept. 2021 - March 2024

- **Thesis:** *Self-Supervised Learning and Model Adaptation for Facial Attribute Classification* (LoRA, PEViT).

University of Calabria

Arcavacata (CS), Italy

B.S. IN COMPUTER SCIENCE (GRADE: 108/110)

Sept. 2018 - Sept. 2021